

Lead AI & ML Engineer | Production ML & LLM Systems

Lead AI and ML engineer with 10+ years of engineering experience building production ML and LLM systems. Lead reusable fine-tuning and serving infrastructure at Best Buy, with prior GPU-accelerated cybersecurity pipelines and Azure ML delivery. Work spans model evaluation, deployment, monitoring, and retraining, alongside technical mentoring and cross-functional delivery.

Skills

- **LLM Training & Serving:** LLM Fine-tuning; LoRA / QLoRA; Distributed Training; LLM Evaluation; OpenAI-Compatible Serving; Multi-Task Adapters
 - **Modelling & Analytical Methods:** Classical ML; Deep Learning; Computer Vision; Natural Language Processing (NLP); Time-Series Forecasting; Feature Engineering; Behavioral Anomaly Detection; Autoencoders; Operations Research; Vehicle Routing; Mixed-Integer Programming; Constraint Modelling; Supply-Chain Optimization
 - **Frameworks & APIs:** FastAPI; scikit-learn; XGBoost; PyTorch; TensorFlow / Keras; TensorFlow Serving; Hugging Face Transformers / PEFT; Unsloth; vLLM; Google OR-Tools
 - **GPU & Distributed Computing:** CUDA GPU Libraries; GPU-Accelerated Computing; RAPIDS cuDF / CuPy
 - **Experimentation & Model Lifecycle:** MLflow; Experiment Tracking; Model Evaluation; Model Versioning / Registry; Automated Retraining; Vertex AI Pipelines; Azure ML Pipelines; Vertex AI Feature Store
 - **Cloud & Serverless Services:** GCP; Microsoft Azure; GKE; Vertex AI; Azure ML; Cloud Run; Cloud Functions; Azure Function Apps; Logic Apps
 - **Deployment & Production Reliability:** ML System Architecture; Cloud-Native ML Systems; Docker; Kubernetes; Helm; Seldon Core; Istio; HPA Autoscaling; CI/CD; Cloud Build; Azure DevOps / Pipelines; ArgoCD; Google KCC; GitOps; JWT Authentication
 - **Monitoring, Testing & Reporting:** Prometheus; Grafana; Google Cloud Monitoring; Model Monitoring; Load Testing; Unit / Integration Testing; Streamlit; Power BI
 - **Languages, Data & Storage:** Python; SQL; Apache Beam / Dataflow; Pub/Sub; BigQuery; MS SQL Server; Redis Caching; GCS; Azure Blob Storage; Artifact Registry; Pandas / NumPy; Dask; Git; Linux / Shell Scripting
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Experience

Lead Artificial Intelligence Engineer

Best Buy | Jul 2025 - Present

- Engineered a configuration-driven **LLM fine-tuning framework** supporting distributed training, task-specific adapters, and automated evaluation. Standardized task configuration and model artifacts enabled multiple tasks to share a common serving deployment.
- Deployed **self-hosted LLM serving** with vLLM on GKE and **estimated annual API-cost savings of ~\$300K**, keeping model inputs within company-managed infrastructure and reducing exposure to external API availability and behavior during peak demand.
- Built **production fraud-detection systems** spanning multiple models and batch, streaming, and real-time components, including a feature pipeline handling **40,000 events/sec** at peak. Supported a **200 ms scoring SLA** and improved fraud-score coverage by **~20%** during holiday traffic.
- Lead architecture and production delivery of reusable LLM infrastructure; partner with data scientists to deploy secure, monitored retail AI services.

Lead ML Engineer - ML Engineering & Operations

Wipro | Aug 2023 - Jul 2025

- Engineered **AWS account digital fingerprinting** on NVIDIA DGX infrastructure using GPU-accelerated data processing. Trained per-entity behavioral autoencoders on **billion-row security datasets** to detect anomalies, with continuous retraining and model management.
- Scaled **Proofpoint phishing-detection pipelines to 2,400 emails/sec**, double the previous rate. Incorporated analyst feedback and Microsoft Sentinel integration to identify phishing missed by initial filters and support quarantine response.
- Automated **abuse-inbox triage** with ML routing and OCR extraction across email bodies and attachments. Routed scammer identifiers to security teams and mail partners for takedowns, **doubling the prior case volume** and reducing manual review.
- Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery; screened **300+ resumes** and conducted **50+ technical interviews**. Received Best Buy AML Hall of Fame recognition (2024) during the Wipro client engagement.

Senior Data Scientist - Data Science / Azure ML

Accenture | May 2021 - Sep 2023

- Developed **product-quality prediction for BamaGruppen** on Azure ML, with automated retraining and stakeholder alerts. Increased truck inspections from **130 to 160/day**, defect detection by **17%**, and customer approval by **20%**, delivering **\$2.5M annual savings**.
- Designed **last-mile vehicle routing** with OR-Tools under capacity, delivery-window, waiting-time, and store-demand constraints. Compared optimized routes with manual plans to improve resource utilization and reduce logistics costs.
- Implemented **transport optimization** using mixed-integer assignment models and APIs, automating daily dispatch and replacing manual Excel planning.
- Modelled **supply-chain and distribution-hub scenarios** across cost, freshness, carbon footprint, and warehouse volumes to support network-design decisions.
- Led cross-functional delivery, mentored data scientists, and presented recommendations to Norwegian stakeholders.

Data Scientist

GD Research Centre Pvt Ltd | May 2020 - May 2021

- Developed domain-aware **GlobalData forecasts** and scenario analysis across **15+ industry verticals**, informing market-intelligence reports and resource-allocation decisions.

Deep Learning Engineer

Tata Consultancy Services | Jul 2019 - Mar 2020

- Built and deployed an **end-to-end ML system for semiconductor defect detection**, using distributed training on **~1 million images in ~3 hours** and achieving **90% classification accuracy**. Deployed the system at Amkor in South Korea and trained the client team to operate it.

Senior Software Engineer - ML

Infosys | May 2016 - Jul 2019

- Implemented **opportunity win/lose prediction** with logistic-regression scoring and data pipelines, integrating predictions into business dashboards for sales prioritization.
- Delivered tested **insurance ID-card generation** workflows, from impact analysis and design through implementation, support, and team knowledge sharing.

Open-Source Project

vLLM Model Chooser

Built an interactive GPU-to-model sizing tool with VRAM, quantization, and KV-cache fit filters. [Live project](#).

Certifications

- Google Cloud Certified: Generative AI Leader | Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure Data Scientist Associate | Microsoft Certified: Power BI Data Analyst Associate
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional

Education

B.E. Electrical & Electronics

Andhra University, Visakhapatnam | 2012 - 2016 · 80%